REPORT: MINI PROJECT 2025-26	
PROJECT NAME (Group Number)	**FPGA Based Automatic Irrigation System (Group Number 9)**
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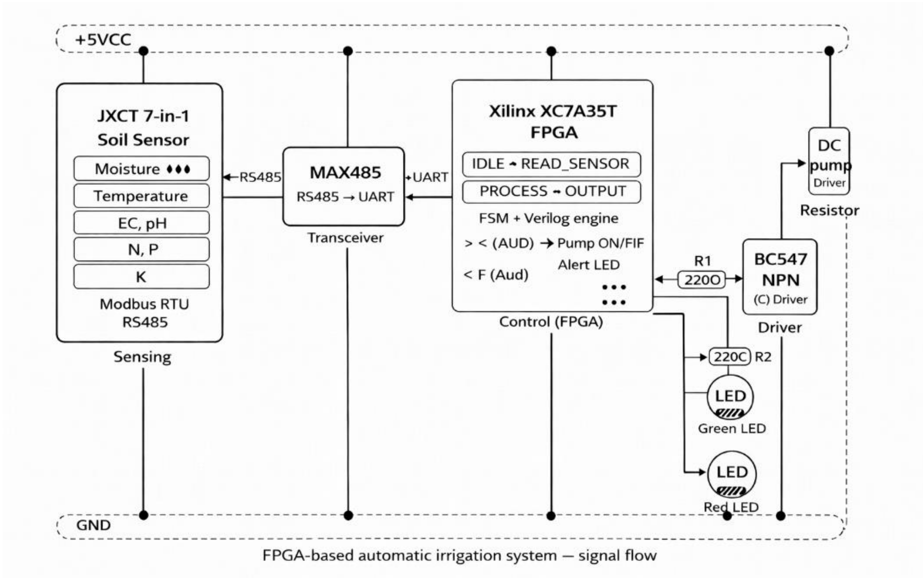
SEM/Year/CAY	VI / Third Year / 2025-26
Domain	Embedded Systems and FPGA-based Digital Control
Project Name	FPGA Based Automatic Irrigation System
Problem Statement (Initial Goal)	Traditional irrigation systems rely only on moisture sensors, ignoring soil salinity, pH, temperature, and nutrient levels, which can cause crop damage and waste 40–60% of water. This project develops a low-cost FPGA-based automatic irrigation system using the JXCT 7-in-1 Soil Sensor (Modbus RS485, 5V) to monitor all seven soil parameters, make real-time intelligent pump control decisions, and generate nutrient deficiency alerts, validated through professional HDL simulation tools.
OBJECTIVE(s)	1) Monitor 7 soil parameters (Moisture, Temperature, EC, pH, N, P, K) simultaneously via Modbus RTU RS485. 2) Implement Verilog HDL Finite State Machine on Xilinx XC7A35T Artix-7 FPGA. 3) Activate pump ONLY when soil is dry & all safety conditions are safe. 4) Generate real-time NPK nutrient deficiency alert signal. 5) Validate design using EDA Playground, Tinkercad, and FPGA web simulator.
SPECIFIC	To build an intelligent FPGA-based automatic irrigation system using the JXCT 7-in-1 soil sensor (IP68, Modbus RS485, 5V) to monitor moisture, temperature, EC, pH, N, P, and K simultaneously. The Xilinx XC7A35T Artix-7 FPGA, programmed in Verilog HDL, acts as the controller. A 4-state FSM evaluates all parameters using threshold

	comparators and controls the water pump, LEDs, and nutrient alerts through a BC547 transistor circuit. The design is validated using EDA Playground, Tinkercad, and a custom FPGA web simulator.
MEASURABLE	A synthesizable Verilog HDL irrigation system will be verified using 6 test cases covering safe irrigation, unsafe conditions (temperature, EC, pH), and nutrient deficiency alerts. EPWave and the custom web simulator will confirm correct signal behavior. All 6 test cases target 100% accuracy.
ACHIEVABLE	The project uses EDA Playground, Xilinx Vivado, Tinkercad, and a custom HTML/JS FPGA simulator. Hardware includes XC7A35T FPGA, MAX485, BC547, resistors, 5V pump, LEDs, and JXCT 7-in-1 sensor. Sensor procurement will be completed upon receipt of pending funding.
RELEVANT	The prototype supports precision agriculture and water conservation schemes. Unlike microcontroller systems, FPGA provides deterministic parallel hardware control. The 7-parameter logic prevents unsafe irrigation while generating nutrient alerts. Future upgrades may include IoT, solar power, mobile control, multi-zone irrigation, and AI-based predictive irrigation.
TIME-BOUND	Week 1-2 (Jan 2026): System architecture design; Verilog HDL module declaration, FSM design, threshold constants definition. Week 3-4 (Feb 2026): Complete Verilog coding; testbench writing; EDA Playground simulation with Icarus Verilog; EPWave waveform verification of all 6 test cases. Week 5 (Mar 2026): Tinkercad circuit simulation with transistor, motor, LED; custom FPGA web simulator development and mobile compatibility. Week 6-7 (Apr-May 2026): Hardware procurement (JXCT sensor); MAX485 RS485 interface; Xilinx Vivado synthesis and bitstream upload to XC7A35T board. Week 8 (May 2026): Field soil testing; PCB design; final project report writing, documentation, and viva preparation.
INTRODUCTION	Conventional irrigation systems rely only on moisture sensing and ignore critical soil factors such as salinity, pH, temperature, and nutrients. This project develops an FPGA-based automatic irrigation system using the JXCT 7-in-1 soil sensor and Xilinx XC7A35T FPGA. Using Verilog HDL FSM control, the system provides deterministic hardware-level response, enabling water saving and crop protection through multi-parameter safety logic.
DESCRIPTION	The system has four layers: sensing, communication, processing, and output. The JXCT 7-in-1 sensor measures seven soil parameters and sends data through Modbus RS485. A MAX485 converts RS485 signals for FPGA input. The XC7A35T FPGA runs a 4-state FSM (IDLE, READ_SENSOR, PROCESS, OUTPUT) to evaluate thresholds and control the water pump, green/red LEDs, and alert LED through a BC547 transistor. Validation is performed using EDA Playground, Tinkercad, and a custom FPGA web simulator, with all test cases showing correct output.

BLOCK DIAGRAM

System Block Diagram — FPGA Based Automatic Irrigation System



Signal Flow: JXCT Sensor → RS485 A+/B– → MAX485 → UART RX → FPGA FSM

FPGA P7 → R1 (220Ω) → BC547 Base → Collector → Pump (−) | Pump (+) → VCC

FPGA P6 → R2 (220Ω) → Green LED | FPGA P5 → R3 (220Ω) → Alert LED

Mentor Name & Signature with date